Dr. Madhusudhanan Srinivasan

Scientist · Visualization and Machine Learning

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"The purpose of computing is insight, not numbers" – Richard Hamming

Summary

Lead Scientist and **Founding Staff** of one of the largest state-of-the-art Visualization Core Laboratory at King Abdullah University of Science and Technology (KAUST). Staff Scientist at HPE Cray, pioneers of the first US Exascale systems. High Performance Computing (HPC) Visualization and Machine Learning professional with a strong expertise and over 12 years of experience in applied-research, training, and building, operating scalable workflows for visualization and large-data analysis.

Work Experience

HPE Cray

Seattle, WA, USA

Lead Scientist, Visualization and Machine Learning

September 2019 - Present

- Provide technical leadership within Cray's Alexandria Center of Excellence (COE). Analyze, understand, and visualize vast amounts of data proprietary to Cray's High Performance Computing systems. Act as the point of contact for the Alexandria Center for Excellence team throughout Cray.
- Leverage advanced Visualization and Machine Learning techniques to help Subject Matter Experts (SMEs) within Cray define, explore and predict systems behavior and failures. Conduct research on the data that Cray systems produce in order to increase system and application performance.
- Integrate open source analytics and AI projects, write application code, understand data science workflows, and troubleshoot and debug distributed systems. Manage multiple projects and tasks, prioritize work, and develop software projects on time.

KAUST Visualization Core Lab

Thuwal, Saudi Arabia

Lead Scientist

July 2014 - July 2019

- Lead and direct scientific projects for large data visualization and data analysis. Collaborate with and provide support to the University's scientific communities and regional institutions. Manage visualization facilities, mentor and develop a team of 6 visualization and data science experts.
- **Scientific Projects:** Enable scientific discovery through HPC Visualization and Data Analysis workflows in the areas of Clean Combustion Science, Marine Ecology, Earth and Climate Sciences, and Computational Fluid Dynamics (CFD).
- **Facilities:** Build and manage operations for large-scale visualization facilities such as Display Walls and CAVE Virtual Environments. The facilities represent a capital investment of around $10 million, with an annual operational budget of $1 million.
- **Project Management:** Collaborate with scientific users to gather requirements, define scope, schedule and track project milestones with my team. Follow agile project principles for providing flexible and responsive solutions to our users' unique scientific challenges.
- **Training and Scientific Outreach:** Design and administer comprehensive training sessions (5-6 annually) to train around 150-200 researchers in the field of visualization and scientific data analysis. Use visualization as a tool to explain scientific research to over 1500 visitors annually.

KAUST Visualization Core Lab

Thuwal, Saudi Arabia

Visualization Scientist

April 2009 - June 2014

- **Scientific Projects:** Led visualization projects to build cutting-edge scientific decision-making tools for domain scientists and industry partners.
- **Software Development:** Architected solutions for large scale data visualization and interaction with distributed graphics on ultra-high resolution tiled-displays and virtual reality environments.
- **HPC Visualization:** Led efforts to design and implement *in situ* and streaming High Performance Computing (HPC) Visualization from supercomputing facilities at KAUST.
- **Training:** Trained and Mentored over 200 researchers annually, in using various tools and frameworks for data visualization.

Laika Studios, Inc.

Portland, OR, USA

Technical Director

August 2008 - December 2008

- Worked with the Front-end Pipeline team to develop tools used by 3D artists and animators in the studio.
- Responsibilities included developing front-end GUI for 3D/2D Digital Asset Management, Maya/Houdini Integration.
- Coordinated daily with production department heads to collect requirements and feature-requests, and provide solutions around their established workflows, using agile principles.
- Negotiated project scope and feature additions, debugged production-related technical problems, supported 3D artists with various aspects of the production pipeline.

Google, Inc.

Mountain View, CA, USA

Engineering Research Intern

January 2007 - March 2007

- Researched various statistical models for enhancing resolution of photographic images.
- Implemented "Statistical Image Super-resolution" using OpenCV and MapReduce technologies.

Skills

Programming Languages	Python, Rust, Javascript, Elixir, C/C++
Data Visualization and 3D Graphics	D3js, Highcharts/AMCharts, Paraview/VTK, Avizo3D, Bokeh, HoloViews, Blender 3D, Unity3D Engine
Web and Databases	Node.js, Phoenix Framework, React.js, HTML5/WebGL, Webpack, OmniSciDB Core
Distributed Computing	MPI, DisplayCluster, Erlang OTP, Paraview/Catalyst, Dask
Data Science	Anaconda, Pandas, SciKit-Learn, Jupyter Lab

Articles and Publications

- 4.2 Workflow Specification. M. Parashar, P. Bremer, K. Heitmann, M. Larsen, J. Patchett, T. Peterka, **M. Srinivasan**, N. Röber, S. Frey, and B. Raffin, In Situ Visualization for Computational Science, Dagstuhl Report vol. 8, no. 7, p. 13, 2019.
- 4.10 Software Complexity, Heterogeneity, and User-facing Issues. E. Bethel, J. Patchett, P. Messmer, M. Parashar, A. Bauer, **M. Srinivasan**, K. Ono, N. Röber, N. Gauger, J. Bennett, et al., In Situ Visualization for Computational Science, Dagstuhl Report vol. 8, no. 7, p. 34, 2019.
- RedSeaAtlas: A Visual Analytics Tool for Spatiotemporal Multivariate Red Sea Data. S. Afzal, S. Ghani, G. Tissington, S. Langodan, T. Jamil, **M. Srinivasan**, and I. Hoteit. Submitted to the Workshop on Visualization in Environmental Sciences 2019 (EnvirVis 2019)
- Sonification of Animal Tracks as an Alternative Representation of Multi-Dimensional Data: A Northern Elephant Seal Example. Duarte CM, Riker P, **Srinivasan M**, Robinson PW, Gallo-Reynoso JP, et al. (2018).Frontiers in Marine Science.
- Simulation and Visualization of the Cyclonic Storm Chapala over the Arabian Sea: A Case Study. T. Theußl, H. P. Dasari, I. Hoteit and **M. Srinivasan**, 4th Saudi International Conference on Information Technology (IEEE Big Data Analysis) (KACSTIT), Riyadh 2016.
- Three-dimensional immersive virtual reality for studying cellular compartments in 3D models from EM preparations of neural tissues. Corrado Cali, Daniya Boges, Jumana Baghabra, Glendon Holst, Anna Kreshuk, Fred Hamprecht, **Madhusudhanan Srinivasan**, Heikki Lehvaslaiho, Pierre Magistretti. Journal of Comparative Neurology, 2015.
- PathlinesExplorer - Image-based exploration of large-scale pathline fields. O. H. Nagoor, M. Hadwiger and **M. Srinivasan**. 2015 IEEE Scientific Visualization Conference (SciVis), Chicago, IL, 2015, pp. 159-160 (Video Preview)
- Query-driven Parallel Exploration of Large Datasets. Atanas Atanasov, **Madhusudhanan Srinivasan**, Tobias Weinzierl, In proceedings of IEEE Symposium on Large-Scale Data Analysis and Visualization, Seattle, WA (2012).
- Semi-automatic end-user tools for construction of virtual avatar behaviors. **Madhusudhanan Srinivasan** and Ronald A. Metoyer, In Proceedings of the 16th International Conference on 3D Web Technology (Web3D 2011). ACM, New York, NY, USA, 121-128. (Video Preview)
- Acquisition of stereo panoramas for display in VR environments. Richard A. Ainsworth, Daniel J. Sandin, Jurgen P. Schulze, Andrew Prudhomme, Thomas A. DeFanti and **Madhusudhanan Srinivasan**, Proc. SPIE 7864, 786416 (2011).
- The Future of the CAVE. DeFanti, T.A., Acevedo, D., Ainsworth, R.A., Brown, M.D., Cutchin, S., Dawe, G., Doerr, K., Johnson, A.E., Knox, C. Kooima, R., Kuester, F. Leigh, J., Long, L., Otto, P., Petrovic, V., Ponto, K., Prudhomme, A., Rao, R., Renambot, L., Sandin, D.J., Schulze, J.P., Smarr, L., **Srinivasan, M.**, Weber, P., Wickham, G. Central European Journal of Engineering 2011, Pages 16-37.
- Towards a Digital Makkah - Using Immersive 3D Environments to Train and Prepare Pilgrims. Jens Schneider, Dina Garatly, **Madhusudhanan Srinivasan**, Stephen Guy, Sean Curtis, Steven Cutchin, Dinesh Manocha, Ming Lin, Alyn Rockwood. International Conference on Digital Media and its Applications in Cultural Heritage, March 2011.
- Multisource Reverse-Time Migration and Full-Waveform Inversion on a GPGPU. C. Boonyasiriwat, G. Zhan, M. Hadwiger, **M.Srinivasan**, and G. T. Schuster. 72nd EAGE Conference and Exhibition incorporating SPE EUROPEC 2010
- Controllable real-time locomotion using mobility maps. **Madhusudhanan Srinivasan**, Ronald A. Metoyer, and Eric N. Mortensen. In Proceedings of Graphics Interface 2005 (GI '05). Canadian Human-Computer Communications Society, School of Computer Science, University of Waterloo, Waterloo, Ontario, Canada, 51-59.
- A Data-Driven Model of Pedestrian Movement. 3rd International Conference on Pedestrian and Evacuation Dynamics. Casburn, L., **Srinivasan M.**, Metoyer, R., and Quinn, M. Vienna, Austria. pp. 189-200
- A Tangible Interface for High-Level Direction of Multiple Animated Characters. Metoyer, R. A.; Xu, L. and **Srinivasan, M.**. In Proceedings of Graphics Interface 2003, pp. 167-176.
- A Responsiveness Metric for Controllable Characters. **M. Srinivasan** and R. Metoyer, School of Electrical Engineering and Computer Science, Oregon State University, Technical Report #CS05-50-01.

Awards, Distinctions & Invited Talks

Dagstuhl Seminar: *In Situ* Visualization for Computational Science — *Dagstuhl, Germany*

Invited Contributor & Participant — *July 2018*

- Group Co-Leader: *Workflow Specification Group*
- Technical Talk: *Supporting the Virtual Red Sea Project at KAUST*
- Group Contributor:
  - *Exascale Requires Different Data*
  - *Software Complexity, Heterogeneity, and User-facing Issues*

Schloss Dagstuhl, a nonprofit institution with associates from eleven universities and research organizations, is one of the world's leading research centers in informatics. It has been hosting invitation-only Dagstuhl Seminars and Dagstuhl Perspectives Workshops since the center's founding in

1990. The seminar workshop on *In Situ* Visualization for Computational Science **hosted 35 world-leading experts to identify ten challenges for in-situ visualization and processing that require significant research.**

In Situ Visualization, i.e., visualizing simulation data as it is generated, is an emerging processing paradigm in response to trends in the area of high-performance computing. This paradigm holds great promise in its ability to access increased spatio-temporal resolution and leverage extensive computational power. However, the paradigm is also widely viewed as limiting when it comes to exploration-oriented use cases and further will require visualization systems to become more and more complicated and constrained. Additionally, there are many open research topics with *in situ* visualization. The seminar brought together researchers and practitioners from three communities (computational science, high-performance computing, and scientific visualization) to share interesting findings, to identify lines of open research, and to determine a medium-term research agenda that addresses the most pressing problems. These challenges were identified by spending the first day of the workshop with participants giving short presentations on their experiences with in-situ processing, with a special focus on unsolved problems. The participant perspectives were then organized into the ten research challenges. Over the following days, sub-groups discussed each of the ten challenges and then presented the key points of their discussions to the group and received feedback. Shortly after the workshop, the leaders of each sub-group wrote summaries for its associated research challenge; these summaries are the basis of the report In Situ Visualization for Computational Science (Dagstuhl Seminar 18271), Volume 8, Issue 7, July 2018.

Leading@KAUST Program by Harvard Business Publishing

KAUST, Saudi Arabia

Nominated Participant

March 2015

The Leading@KAUST Management program administered by Harvard Business Publishing is an exclusive program for nominated high-performing staff that follows a blended learning approach for those transitioning into a leadership role. A group of 25 distinguished staff were selected for a 10-week program. The program elements are designed to

- Ensure a smooth transitional experience to supervisory/managerial roles
- Enforce the leadership and managerial skills of our managers to lead and coach their teams effectively
- Enhance existing knowledge and acquire new, in relation to contemporary management best practices

Workshop on Visualization Infrastructure & Systems Technology

New Orleans, U.S.A

Invited Speaker: KAUST Visualization Laboratory - Sustaining and Operating at Scale

November 2014

The VISTech (Visualization Infrastructure & Systems Technology) workshop is organized as part of the Supercomputing Conference, and explores the intersection between human perception and large-scale visual analysis through the study of visualization interfaces and interactive displays. This rich intersection includes virtual reality systems, visualization through augmented reality, large-scale visualization systems, novel visualization interfaces, high-resolution interfaces, mobile displays, and visualization display middleware. The workshop provides a forum in which experts in the large-scale visualization technology field and users can come together to discuss state-of-the art technologies for visualization infrastructures and systems.

Recognition for Distinguished Service

KAUST, Saudi Arabia

Five-year Service Award

May 2014

Presented by the President of the University, in recognition and appreciation of five years of loyal and distinguished service with King Abdullah University of Science and Technology.

Microsoft Research eScience Workshop

Stockholm, Sweden

Invited Panel Speaker: *Is NUI "Natural" for Scientists?*

December 2011

Each year, 150 top researchers join with Microsoft Research to examine the successes and opportunities for research in the field of data-intensive science. The goal of this eighth annual cross-disciplinary workshop is to bring together scientists from diverse disciplines to share their research and discuss how computing is transforming their work.

Natural user interface (NUI) technology has captured imaginations across industry, education, and entertainment with the release of groundbreaking new technologies for interacting with computer systems, such as Microsoft Surface and Kinect for Xbox 360. Technology is becoming more natural and intuitive - people use touch and speech to interact with their phones, at the ATM, at the grocery store, and in their cars. The learning curve for working with computers is becoming less and less of a barrier, thanks to more natural ways to interact. How is this technology being used by scientists today, and what is its potential? This panel explores best practices and new directions, ultimately seeking to present a slice through the current research agenda for NUI research in scientific applications.

Program Committees

2019	**Program Committee**, EuroGraphics Symposium on Parallel Graphics and Visualization	*Porto, Portugal*
2018	**Program Committee**, EuroGraphics Symposium on Parallel Graphics and Visualization	*Brno, Czech*
2017	**Program Committee**, EuroGraphics Symposium on Parallel Graphics and Visualization	*Barcelona, Spain*
2016	**Program Committee**, IEEE Symposium on Large Data Analysis and Visualization	*Baltimore, MD*
2014	**Program Committee**, IEEE Symposium on Large Data Analysis and Visualization	*Paris, France*
2013	**Program Committee**, IEEE Symposium on Large Data Analysis and Visualization	*Atlanta, GA*
2017	**Organizing Committee**, High Performance Computing - Saudi Arabia	*Saudi Arabia*
2015	**Planning Committee**, Supercomputing Conference	*Austin, TX*
2014	**Planning Committee**, Supercomputing Conference	*New Orleans, LA*
2015	**Co-Organizer**, Workshop on Visualization Infrastructure and Systems Technology	*Austin, TX*

EuroGraphics Symposium on Parallel Graphics and Visualization: The aim of this symposium is to foster the exchange of experiences and knowledge exploiting and defining new trends in parallel graphics and visualization. The importance of parallel computing is increasing rapidly with the ubiquitous availability of multi-core CPUs, GPUs, and cluster systems. Computationally demanding and data-intensive applications in graphics and visualization are strongly affected by this trend and require novel, efficient parallel solutions.

IEEE Symposium on Large Data Analysis and Visualization: Modern large-scale scientific simulations, sensor networks, and experiments are generating enormous datasets, with some projects approaching the multiple exabyte range in the near term. Managing and analyzing large datasets in order to transform it into insight is critical for a variety of disciplines including climate science, nuclear physics, security, materials design, transportation, and urban planning. The tools and approaches needed to mine, analyze, and visualize data at extreme scales can be fully realized only with end-to-end solutions, which demands collective, interdisciplinary efforts. The LDAV symposium brings together domain scientists, data analysts, visualization researchers, users, designers and artists, to foster common ground for solving both near and long-term problems.

High Performance Computing Saudi Arabia: High Performance Computing Saudi Arabia (HPC Saudi) is the premier regional event in the field, where participants can meet each other, share ideas and experiences, and discuss cooperation and collaboration. The event focuses on coordinated efforts for the advancement of the HPC ecosystem in the Kingdom.

Supercomputing Conference Series: The Supercomputing Conference prides itself as one of the most competitive and valued technical conferences in the field of High Performance Computing. It is often listed at the very top of various national conference ranking systems, showcasing the impact that publishing at SC has on the visibility and careers of researchers around the world. Research published at SC often has amongst the highest citation counts and is regularly considered to be the most impactful research in the community.

Visualization Infrastructure & Systems Technology: The workshop is organized as part of the Supercomputing Conference, and explores the intersection between human perception and large-scale visual analysis through the study of visualization interfaces and interactive displays. This rich intersection includes virtual reality systems, visualization through augmented reality, large-scale visualization systems, novel visualization interfaces, high-resolution interfaces, mobile displays, and visualization display middleware. The workshop provides a forum in which experts in the large-scale visualization technology field and users can come together to discuss state-of-the art technologies for visualization infrastructures and systems.

Mentoring and Advising

Thesis Advising

Ph.D. Thesis: Mr. T. Muralidharan — *MGR Research Institute, India*

Hybrid Extensive Machine Learning Algorithm For Domain Specific Features To Detect And Classify Web Spam — *March 2019*

Abstract: Machine learning deals with making machines learn by experience. It involves data science, artificial intelligence and statistics. It is one of today's most rapidly growing technical fields. This thesis presents a Hybrid Extensive Machine Learning Algorithm (HEMLA) for classification and clustering techniques in data mining. Recent progress in machine learning has been driven both by the development of new learning algorithms and theory and by the ongoing explosion in the availability of online data and low-cost computation. Through experimental study it is found that the proposed Hybrid Extensive Machine Learning Algorithm outperforms the existing supervised and unsupervised machine learning algorithms for detecting and classifying Web Spam.

Master's Thesis: Ms. Omniah Nagoor — *KAUST, Saudi Arabia*

Image-based Exploration of Large-Scale Pathline Fields — *May 2014*

Abstract: *PathlinesExplorer* is a novel image-based tool, which has been designed to visualize large-scale pathline fields on a single computer. PathlinesExplorer integrates explorable images (EI) technique with order-independent transparency (OIT) method. The method is unique in that it allows users to handle large data on a single workstation. Although it is a view-dependent method, *PathlinesExplorer* combines both exploration and modification of visual aspects without re-accessing the original huge data. Our approach is based on constructing a per-pixel linked list data structure in which each pixel contains a list of pathline segments. With this view-dependent method, it is possible to filter, color-code, and explore large-scale flow

data in real-time. In addition, optimization techniques such as early-ray termination and deferred shading are applied, which further improves the performance and scalability of this approach.

Master's Thesis: Ms. Roba Binyahib

KAUST, Saudi Arabia

Image-based Exploration of Iso-surfaces for Large Multi-Variable Datasets using Parameter Space

May 2013

Abstract: Explorable Images (EI) are a novel approach that provides limited interactive visualization without the need to re-render from the original data. In this work, the concept of EI has been used to create a workflow that deals with explorable iso-surfaces for scalar fields in a multivariate, time-varying dataset. This work presents a tool that allows a user, at run-time, to interactively browse and calculate a combination of iso-surfaces superimposed on each other. The result is the same as calculating multiple iso-surfaces from the original data but without the memory and processing overhead. The tool also allows the user to change the (scalar) values superimposed on each of the surfaces, modify their color map, and interactively re-light the surfaces. The work demonstrates the effectiveness of this approach over a multi-terabyte combustion dataset. It also illustrates the efficiency and accuracy of the technique by comparing results with those from a more traditional visualization pipeline.

Training & Mentoring

Research Mentor

KAUST, Saudi Arabia

Saudi Research Science Institute Program

July 2018

Research Mentor

KAUST, Saudi Arabia

Saudi Research Science Institute Program

July 2016

The Saudi Research Science Institute Program (SRSI) is a research program that introduces students nation-wide to the transformative potential of science, technology, engineering and math (the STEM fields), and inspires them to pursue further education and careers in these fields. It is the first STEM summer program for high school students in Saudi Arabia to combine theoretical coursework and research. SRSI is a gateway to future studies at KAUST, and is aimed at developing the next generation of scientists, researchers and leaders for Saudi Arabia. Additional details about the program can be found here - SRSI Program Brochure.

Training: HPC Visualization Tools

Jeddah, Saudi Arabia

High Performance Computing – Saudi Arabia

March 2017

HPC Saudi Arabia is a premiere national and regional event in the area of High Performance Computing. The training program at the event provided new high performance computing users in Saudi Arabia and the region with general knowledge on accessing Shaheen (KAUST Supercomputer), including compilers, libraries, software available to users and using examples to schedule and run efficiently the desired application. Specifically, a training on examples and use of Visualization and analysis tools on Shaheen was provided to over 120 participants from Saudi Arabia and the surrounding region.

Education

Ph.D. Computer Science

July 2005 - March 2009

Oregon State University, College of Engineering

Oregon, USA

GPA 3.91/4.00

Doctoral Dissertation: Behavior graphs for data-driven animation of 3D characters

This dissertation presents user-centric tools and techniques for the design of an interactive motion data-structure to animate 3D virtual avatars. The approach automatically analyzes a traditional labeled motion graph and constructs a condensed, coarser graph referred to as a Behavior Finite State Machine (BFSM) using reinforcement learning techniques. Each node of the BFSM represents a single behavior, and each weighted edge represents a valid transition and its likelihood. A semi-automated mechanism is presented for editing the graph by a user. The work also introduces an interface for user-in-the-loop design of transitions and blend-spaces to generate controllable avatars from a BFSM. The results of our approach using two dynamic interactive examples, locomotion and martial arts, are presented in this work.

Relevant Skills: Multi-dimensional Data Analysis, Graph Algorithms, Reinforcement Learning, 3D Computer Graphics, 3D Visualization

M.S. Computer Science

September 2001 - June 2005

Oregon State University, College of Engineering

Oregon, USA

GPA 3.90/4.00

Master's Thesis: Interactive human locomotion using motion graphs and mobility maps

Graph-based approaches for sequencing motion capture data have produced some of the most realistic and controllable character motion to date. Most previous graph-based approaches have employed a run-time global search to find paths through the motion graph that meet user-defined constraints such as a desired locomotion path. Such searches do not scale well to large numbers of characters. In this work, we describe a locomotion

approach that benefits from the realism of graph-based approaches while maintaining basic user control and scaling well to large numbers of characters. Our approach is based on precomputing multiple least cost sequences from every state in a state-action graph. We store these precomputed sequences in a data structure called a mobility map and perform a local search of this map at run-time to generate motion sequences in real time that achieve user constraints in a natural manner. We demonstrate the quality of the motion through various example locomotion tasks including target tracking and collision avoidance. We demonstrate scalability by animating crowds of up to 150 rendered articulated walking characters at real-time rates.

Relevant Skills: Data Mining, Graph Algorithms, 3D Computer Graphics, 3D Animation

B.Engg. Computer Science

July 1997 - May 2001

Maharaja Sayajirao University, Faculty of Engineering and Technology

Baroda, India

GPA 3.99/4.00. First Class with Distinction